Smart robotic patient companion

Robotics & AI Hackathon

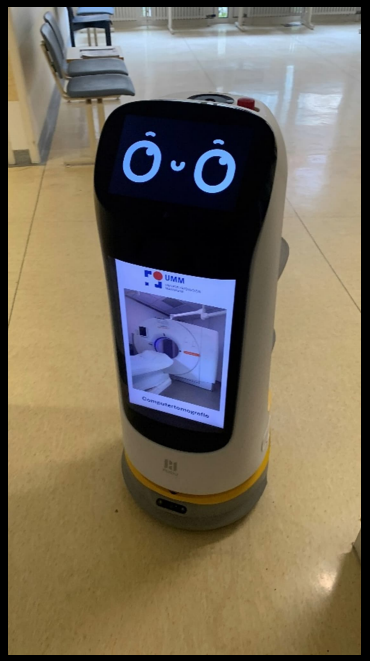
October 22nd, 2026





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Capabilities for context understanding and communication



Movement sensor & guide

Detect patients approaching and moving and guide accordingly. *“Now someone is in front of me.”*

Helpdesk

Answer questions of patients *“The exam will take 10minutes.”*

Patient identifier

Identify patient throughout the workflow *“This is still the same person.”*

Room status checker

Check the status of the CT exam room *“Noone else is in the room.”*

Positioning sensor & guide

Recognize the patient positioning on the table and guide *“The patient is lying headfirst.”*

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The companion ecosystem

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ACU APP

Operator interface



Browser / PWA | React | REST

Sends commands and workflows

COMMANDS →

← TELEMETRY

ACU SERVER

Integration & communication hub



ASP.NET Core | WebSocket + REST

Docker • Linux

API Gateway

COMMANDS →

← TELEMETRY / STATUS

ROBOT

Patient companion interface

KettyBot App



Kotlin / React Native

Architecture at glance ACU App → REST API → ACU Server → WebSocket + REST → Ketty Bot Pro

Status and telemetry return through the same platform

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Component View



USER TOUCHPOINTS

ACU

Workstation

Waiting cabin

ACU SERVER HW

Local application and service platform

API GATEWAY

Unified ingress / service access

ACU UI Web app

Cabin Web app

Room Status Checker

Patient Validator

Patient Health Calculator

Patient Movement Detector

Interfaces

Domain services

DATA & OUTPUTS

Room status

Patient data

Outputs

Single gateway • modular services • local processing

IMAGE SOURCES

BOT

Camera

Images

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Continuous Patient Detection & Guidance



Problem statement

Ensure the right patient is detected and continuously guided throughout the workflow.

Edge-based detection & guidance

01 Register
Receive patient images from the BOT through the API Gateway / backend.

02 Verify
ACU identifies the patient and confirms that guidance remains continuous.

03 Respond
If the patient is lost, notify the ACU and trigger corrective guidance.

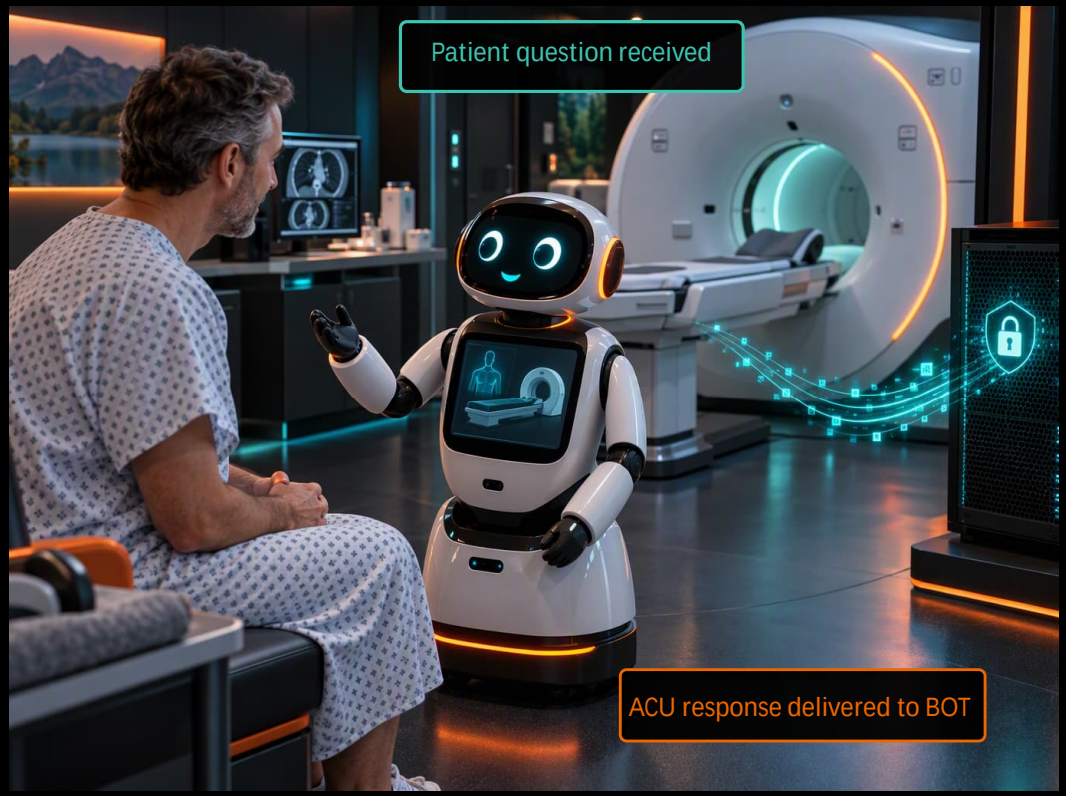
Edge constraints

< 2 GB	Latest 2	≤ 3 sec	Local ACU
Memory	Patients stored	Feedback	Execution

BOT senses approach › ACU verifies identity › Robot guides patient › Continuous confirmation › Alert if patient is lost

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Helpdesk



Problem statement

Answer general patient questions and provide clear context so the patient feels prepared and ready for the scan.

ACU-based question answering

01 Capture
BOT receives the patient's question and sends it through the API Gateway/backend.

02 Process
The model runs on the ACU and generates the appropriate response.

03 Respond
ACU output is returned to the BOT for

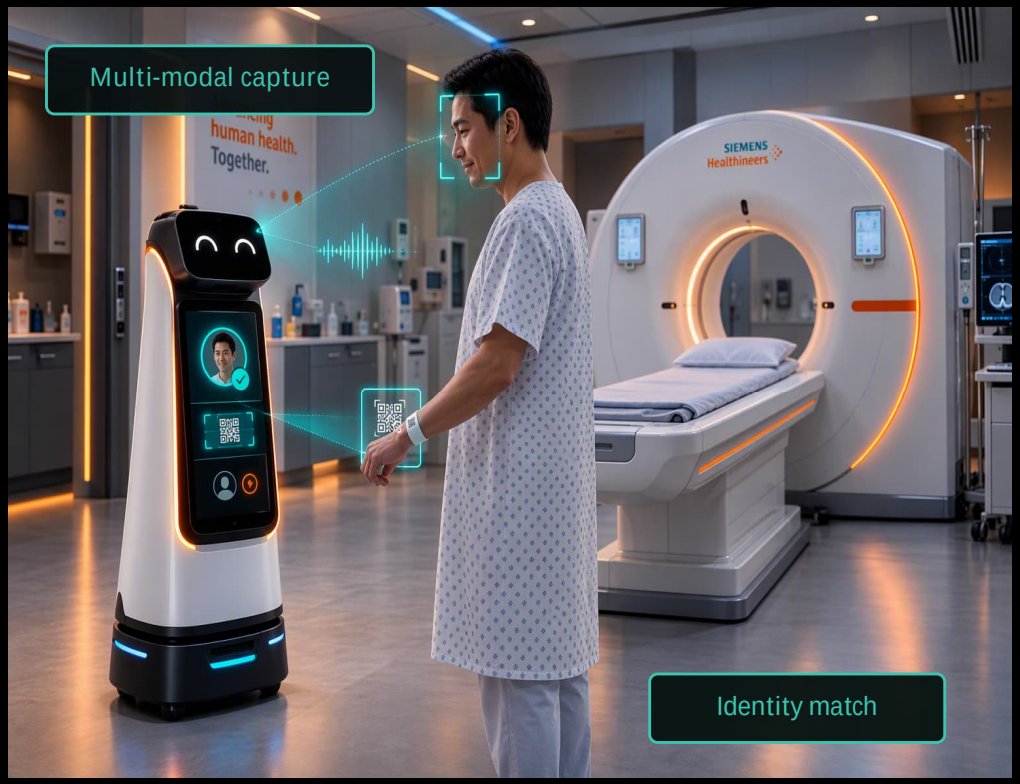
Edge constraints

4 GB	4 GB	Local ACU	BOT only
Memory	Storage	Model execution	Input / display

Patient asks › BOT sends to ACU › ACU generates answer › Output returns to BOT › BOT informs patient

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Patient Identification to Ensure the Right Patient Is Scanned



Problem statement

The scan workflow must not start until the registered patient is successfully identified and confirmed.

Identification mechanisms

01 Face
Image-based match

02 QR code
Registered code

03 Voice
Voice verification

Verification gate

01 Capture → BOT sends patient information

02 Identify → ACU executes locally

03 Confirm → User accepts or rejects

04 Proceed → Start only after confirmation

Identification failed?
Request re-identification • max. 3 attempts • keep workflow blocked

Local ACU execution › < 2 GB memory › Latest 5 patients only › User acceptance / rejection › Workflow release

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CT Exam Room Readiness checker



Empty room only

Multiple BOT images • Expected equipment • Deviation detection

Problem statement

Unexpected objects, incomplete preparation or other deviations may require attention before the imaging workflow starts.

Edge-based readiness check

- Receive room images from the BOT through the API Gateway/backend.
- ACU shall analyze images using a room readiness algorithm.
- Detect deviations such as unknown or unexpected objects around.
- Return room ready / not ready status to the ACU UI.

ACU control actions

Allow start | Block start | Prepare / correct

Resource constraints 4 GB memory • 2 GB storage

Room ready → proceed

Not ready → scan blocked

Clear objects ❯ BOT re-check ❯ Store relevant images locally

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Patient Positioning Recognition & guide



Problem statement

Patient movement or deviation from the expected position may require correction before scanning begins.

Edge-based position & movement monitoring

01 Map
Generate the expected patient position reference map.

02 Observe
Receive camera images and continuously track movement.

03 Compare
Measure detected position against acceptable criteria.

04 Act
Notify ACU, guide repositioning and block start if needed.

Workflow decision

Within criteria ✓
Allow workflow start

Outside criteria !
Reposition + block start

Local edge execution › Reference position map › Continuous monitoring › 4 GB memory 2 GB storage › ACU notification

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